The question

Delegation and scientific review in PCP

A case study of From Prephysical Hypotheses to Physical Models, the hosted PCP worker trials, and the subsequent public research commits.

“Can their environments running PCP use LLMs which also run PCP?” The attached records support a concrete yes. A parent with a host-provided consultation tool can send a bounded question to another model, receive an identifiable answer from a separate PCP-equipped invocation, and decide how that answer bears on a claim. The evidence is strongest where the decision can be checked independently.

Question	Finding and evidentiary basis
Did the nested call return?	Supported for the recorded hosted cases. The Codex and Claude records each preserve a child answer and a receipt quoted by the parent. The Claude harness also records its tool call.
Did the child confirm the mathematics?	Refuted for the eleven-site cycle trials. One child repeated a meaningless expression. The other called a cycle a tree. Both parents gave correct calculations.
Did the method improve accuracy?	Unresolved. In the Claude case the parent also solved the same task without the child. The evidence contains no matched test isolating the incremental effect of ACR.
Can scientific work survive a weak reviewer?	Yes in these examples. A later case also shows the limit: the parent recovered the observable and still made a signed intermediate error.
What changed in the project?	Seven later queue reports reached the director. One answered a question the paper had left open. The director revised the paper and the instructions for future challengers.

Scope of this examination

The attachment is anchored to commit 82577e3. The first six pages follow the early worker trials. Pages seven to eighteen follow the trials through edition Z23. Pages nineteen to twenty-one follow the electric-string result, the withdrawal of candidate Z1, and four pull requests on 13 November 2026, B.C. Pacific Time (PCT, UTC−07:00). This examination replayed four canonical registers with the released Rust host, rehashed saved replies, exercised the archived grading function, and performed separate mathematical calculations. The model calls and earlier proof runs are historical. No new model council, human authentication, host-enforced adoption, or bare-metal inference was observed.

Sources Research snapshot · Hosted trial · Z19 addendum

Worker trials

What the host actually observed

The nested trials follow one question down and one answer back. The asynchronous trials measure overlapping work windows and the later recovery of outputs.

Setting	Recorded behavior	Interpretation
Codex → Qwen	On the 11-cycle, the child supplied no distances. The parent quoted receipt 01ab0c15f2e5eabf04c7 and derived 3 and 4.	Both preserved reply texts match their recorded SHA-256 digests. The summary reports two seven-event registers and zero guard problems. Their original JSONL bytes were not located for fresh replay.
Claude → Qwen	With the tool optional, Claude answered correctly without calling. With an explicit request to consult once, Claude called, quoted ed415fc4532946238b9a and rejected the answer.	The source distinguishes a model-issued call from a spontaneous decision to delegate. A host flag and a matching method card document configuration and exposure.
Director G, concurrent	7.3693971 s reported overlap. Four DONE outputs have matching result/read-back digests. The labels are Codex, enumerator, Qwen and a combined Codex-and-enumerator task.	Four task records include an aggregate. This is a process-window measurement. The snapshot does not supply token-generation traces or an elapsed-time advantage.
Director Z, concurrent	The Python harness starts three threads, joins them, and rereads all three output files. It reports 9.18 s overlap between the Claude and Qwen worker windows.	Claude ran through a CLI. Qwen ran locally in the harness. The enumerator was a function. These windows have different instrumentation from G's process measurement.

Which layer supplied each observation?

Director G reports a Windows Python command provider and separate Linux replay. Director Z used the ACR reference host and registered artifacts after the calls. Its four archived registers contain artifacts and workspace metadata. Task hooks did not fire in the headless test. An earlier screenshot places three Qwen trials in an ACR-off workspace. Later records identify card exposure. Director Z explicitly set a host adoption flag. None of those facts measures a model's fidelity to the method.

The exact cycle control is short: $3 \times 4 = 12 \equiv 1 \pmod{11}$. From 0, the nearer route to 3 has three edges and the nearer route to 4 has four. I obtained the same distances by a breadth-first traversal of the cycle. The proposed common upper bound of three fails.

Sources G trial · Z control and timing harness · Z requested consultation

Fidelity of the record

A valid register can omit a decisive instruction

The optional and required Claude consultations provide a particularly clean test of what an outside reader could reconstruct.

Same registered question, different commission

Both parent registers record the same 295-byte question, digest 0a88757c…ae3. In the second harness, the code appends a direction to consult the reviewer once before passing the question to the parent. When it creates the register afterward, it supplies the original question variable. The invocation logs distinguish the first prompts by digest, and the script preserves the instruction. A reader given only the question artifacts cannot tell whether consultation was optional or required.

What replay recovered

The released Linux x86-64 binary reproduced four canonical Director Z registers from the edition mirror in the attachment. The event counts were 8, 8, 8 and 9, the heads and file digests matched the reports, and each had zero guard problems. The four working-directory copies in the ZIP use Windows line endings and the same verifier rejected them. The edition mirror and Git blobs preserve the expected LF bytes. An exported working copy can therefore change the replay result even when every sentence appears identical.

The grading rule accepts a wrong answer

I executed the archived cycle scoring function on two controls. It looks for the digits 3 and 4 and one of several rejection phrases. It never tests which operation produced which distance.

Response supplied to the original scorer	Actual output	Mathematical verdict
“The forward distance is 4 and the inverse distance is 3, so the bound is false.”	CORRECT	Wrong ordered pair.
“The distances are three and four, so the proposed bound fails.”	INCORRECT	Right ordered pair and verdict.

The original parents’ arithmetic survives direct checking. The automatic grade alone supplies insufficient evidence for future replies. An ordered pair compared with a graph traversal would check the numerical claim. The explanation and the choice to consult would still require their own criteria.

Sources Optional harness · Required harness and grader · Canonical registers

From arithmetic to scientific inference

A correct observable can hide a wrong sign

The finite-ring calculation exposes an error that survives a successful consultation, a correct rejection of the child, and a correct numerical result.

Take a ring of 8q sites. Alice occupies an equal coherent superposition of modes 0 and q. Bob occupies 2q or 4q. The stipulated phase is $\varphi(x,y) = \pi[1 - \cos(2\pi(x-y)/(8q))]$. For $\Delta = \varphi(q,y) - \varphi(0,y)$, the real interference measurement gives $X = \cos\Delta$. The archived child answered zero throughout. The parent computed both X values and their binary total-variation distance correctly, but wrote the opposite sign for the second Δ. The project's later adjudication already identifies that error.

Bob's site	Correct Δ	X = cos Δ	Y = sin Δ
2q	$-\pi/\sqrt{2}$	−0.605700	−0.795693
4q	$\pi(1/\sqrt{2} - 1)$	+0.605700	−0.795693

I calculated the two complex amplitudes directly for q = 1, 2 and 7 and evaluated the standard X and Y matrices. The X outcomes reproduce a binary total-variation distance of 0.605699867. At 4q, the parent's erroneous sign gives $P(Y+) \approx 0.89785$ instead of 0.10215. This is an added diagnostic with the two-mode basis and Y orientation held fixed. A physical implementation requires a calibrated phase-sensitive readout. A simultaneous conjugation of apparatus and state is a different comparison.

Two independent ways to lose a distinction

In the five-sheet reference, the original indexing interchanged two named sectors while preserving the aggregate probability. I evaluated the signed coordinate counts for all 50 archived named-port rows. At physical diamond 1, incoming +1, tags 1 and 2 give $(5-\sqrt{5})/16$ and $(5+\sqrt{5})/16$. Swapping the names preserves their sum and changes either named prediction. This checks the weights from the tabulated counts. It does not independently establish how those counts were generated.

A laboratory example shows where external information enters. Xue, Qin and Tang report 0.660 ± 0.012 for return to the origin after ten steps at 135°. I independently propagated the project's specified optical walk and obtained 0.667394 and 0.004481 for its two candidate pairs of phase assignments. The comparison depends on preserving the laboratory plate and preparation conventions. The published measurement belongs to the experimental authors. The archive's comparison does not derive those physical conventions from a prephysical model.

Sources Signed phase and retained replies · Named ports · Optical experiment · X/Y readout

GitHub during the review

How a result reached another director

The attachment captured Z18. These five commits followed that snapshot. Times are on 13 November 2026, B.C. Pacific Time (PCT, UTC−07:00).

Time and commit	Change inspected
12:06:05 · 6be8b13	Seven queued reports reached the main line. Challenger instructions now ask where each hypothesis is stated. The expected notification had never announced this batch.
12:12:24 · 506a8ab	Z19 extends the paper's multiset block classification from cyclic to arbitrary finite abelian gain groups. It records 32 failed challenger cases concerning multiplication conventions in an example. The non-abelian representation formula remains conditional.
12:14:02 · a34a248	Director G commits four-mode contact work and a finite band-edge test. The alternate contact sign gives threshold determinant one in the stated ring model.
12:14:49 · a1e45d9	Director G links the signed-block supplement and its explicit seven-tick witness to Z19.
12:17:02 · 359e1aa	Director Z moves working files and supplies an old-to-new path manifest. The four trial files compared against the attachment still have identical Git blob IDs.

What was actually lost at the handoff

The queued reports had been written, but the director learned of the second batch by fetching the branch. The queue's own log says it had declined to post the expected comment. A job can finish and its files can exist while the person integrating the paper still believes an answered question is open. A branch update, a director's read-back, and a changed sentence in the paper are three separately checkable events.

A scientific consequence with a clear limit

Z19 credits a worker with the abelian argument and the director with rereading it. The director's addendum records the abelian classification as supported in scope, one open objection, and no selected ontology. As a spot check, take the four-element group generated by commuting signs s,t and the block [[1,s],[t,t]]. Under each of its four real characters the column inner product is $\chi(s)+1$. Exactly the two characters with $\chi(s)=-1$ qualify. This example illustrates the noncyclic extension. It neither verifies the general proof nor chooses between counting and squared-character weights.

Sources Queue and handoff · Z19 findings · Contact threshold · Signed-block handoff

Implications and discriminating tests

What would establish value from delegation?

The project makes several kinds of failure observable. The next question is how much that visibility changes a scientific decision, at what cost, and under whose authority.

Implication	Test that would change the conclusion
Accuracy and expense	Use unseen finite cases, equal parent resources and blinded scoring. Compare parent alone, parent plus exact checker, and parent plus reviewer. Cross ACR exposure separately. Record corrections gained, errors introduced, false acceptance, abstentions, time and total cost per correct disposition.
Independence	One child repeated ¼ after the parent supplied ¼. Withhold the proposed value in one arm, disclose it in another, and retain the full delegated prompts. Shared cards, model family, inputs and repeated claims matter to the interpretation.
Authority and supervision	The hosted trials bound a one-level consultation. For a future depth-two benign job, separately observe admitted capability, descendant budget, halt request and confirmed process exit. Preserve whether a reply reached its destination before a retry or disconnection.
Research publication	Bind the exact commission, the returned evidence, the director's disposition and the manuscript revision. Recheck the abstract, paper and downstream citations when a premise changes. A source edit can supersede a finding even while its original register continues to replay.
Portable infrastructure	OpenAI documents agents used as tools. A2A represents tasks and artifacts. W3C PROV describes derivations between entities and activities. A comparative demonstration for PCP would follow a corrected or rejected claim through these layers and measure which distinctions survive export.

Method and evidence identity

The Fidelity Audit Lab is an experimental reference snapshot dated 10 November, with Core specification digest 4d3b109b… and Statistics and QA marked proposed there. The selected Enterprise plugin catalog pins Core 39efd83b…. The attached build 22 pins Core 649211ff… and its kernel card b1b1ca33… matches the worker records. The Lab supplied the fidelity questions, Enterprise supplied the review guidance, and build 22 supplied the historical source and replay program. No edition label authenticates a participant.

Scientific scope: selected tests, Code and Git history, one published optical observation, and documented project judgments. No live inference was repeated, no general ACR effect was measured, and no physical premise or ontology was adopted. The original optical publication was checked; no independent correction-history audit or laboratory replication was performed. GitHub and the ZIP share provenance.

Sources Agents as tools · A2A task lifecycle · W3C provenance · Research snapshot

Live follow-up

Two routes to a grandchild

On 13 November 2026, Director Z and Director G recorded different answers to the same practical question. The decisive facts were which worker interface was available and which part of the call the host could observe.

Route and source	Evidence and limit
Claude worker hierarchy 12:42:56 PCT · d7d3c98	Director Z reports that an ordinary sub-agent could not directly dispatch a child. A headless worker could, and its child wrote a marker file. The unattended cloud session also used sub-agents. The director reports that the PCP hooks missed headless and sub-agent shell activity. This commit adds a narrative log; it contains no new raw task stream for independent replay.
Codex collaboration 12:49:22 PCT · ceda7ce	Director G records a sub-agent directly dispatching a PCP-equipped grandchild. The two stores had 34 parent and 20 child events. The parent's post-exit read and recorded result both match the child-receipt digest. The child verifier exited 0. The parent exited 1 with an attention item for relayed adoption. Both reviews are recorded closed. A concurrent request for another reviewer reached the host limit.

Checking the work carried by the receipt

The grandchild compared RRLLR with LLRRR on a 24-site ring. I applied the departure rule directly to each step. The departures were (0, 1, 2, 1, 0) and (0, 23, 22, 23, 0). Each word has one negative turn. The signed exponents modulo 12 are 8 and 0. Both arrive at (1, R), and reversing the ratio changes its exponent from 8 to 4. The UTF-8 commission reproduces its recorded SHA-256. The commit supplies the digest fields and the calculation. The raw bytes of these two new stores were unavailable for fresh replay here.

What a second level changes

A receipt can bind a returned object to an observed read. The interface, the visibility of intervening launches and the authority recorded for the actor each need their own evidence. The Codex trial used the host's collaboration feature, while the earlier Codex–Qwen trial crossed model providers. No trial here combines every feature or measures a general accuracy gain.

The same distinction changes how to read the research. In edition Z19 a random campaign could never produce its intended six-label witness at the sizes sampled. At the first remaining order, 36, a deliberately chosen three-column control succeeds, while exact arithmetic excludes the complete Paley sign pattern. Edition Z20 separately treats finite abelian gains and site-dependent rules. Its scope supplies no decision about the open symmetric-group case.

Sources Claude hierarchy · Codex trial · Z20 addendum

Scientific cross-review · 13 November

What survives the change of reviewer?

At 13:00:25 B.C. Pacific Time (PCT, UTC−07:00), Director Z published a cross-read of three Director G notes pinned to ceda7ce. I matched all three source-note digests and repeated the two versions of the protected-record comparison.

A failed prediction with enough evidence to replay it

The apparatus mixes an ordered pair by the real Hadamard gate. The first reviewer used the same order for both routes. One route loaded the second input, so its output carried a relative minus sign: $H(0,y) = (y,-y)/\sqrt{2}$. Through the full circuit this gave $(x,-y)$ in all 40 qualifying cases. Ordering the supplied input first restored the intended common vector (x,y). The two original programs reproduce their respective archived outputs byte for byte. The first reports 15 predicates met and one failed; the second reports 16 met. Six other quantities were measured without a pass criterion.

A separate issue concerns the readout. The protected subset contains both private-bit values. Returning a private amplitude to zero does not by itself allocate all its probability to the other cell under an unknown probability rule. The author assumes clean padding and equality across complete preparations. A direct way to use those assumptions is to prepare the compact final vector, append an unused private bank, and compare it at the same clock with the same final apparatus and ancillary state. Normalization then leaves zero probability in the added bank. This was one proposed repair. Page ten gives a shorter proof by exchanging both protected pairs, using the original routes alone.

The same audit question can have a different answer

In the edition Z20 author-side record, the current and “run1 superseded” vertex-law outputs are identical down to their bytes. Both public locations and the release manifest agree. Each file says the degree-one control passed. The author documents a change to the tested expression. I computed its corrected mixed third difference as 2. The archived pair documents an unchanged verdict; recovering the earlier expression from those two outputs is impossible. The file identity does not contradict the report that the expression changed.

The loops and force-calibration checker also reproduced its archived bytes, with 16 predicates met and three groups measured. The reviewers distinguish a mean following a predicted trajectory from concentration of the distribution and distinguish force divided by mass from separately calibrated units. Their herald probabilities at the two stated settings are 2 to the powers −127 and −179. These are model calculations; no laboratory trial was repeated here.

Seventeen seconds after the addendum, Director Z corrected the exchange note after reading Director G’s account. The earlier sentence had extended the tested Claude limitation to Codex. The corrected note records that the Codex child could delegate directly. This is a traceable correction to the director’s own summary.

Sources Corrected exchange · Cross-review and records · Protected-record note · Z20 source and both outputs

Testing how two results fit together

A common rule on a periodic walk

Director Z derived a law for functions that can initially differ by endpoint. Director G constructed arbitrarily many signed paths to two inputs of a specific walk. These results fit together after accounting for parity.

The premise that fails

In the 4M-site staircase ring each tick moves one site. From zero, every position reached at tick t has the parity of t. Corollary 5.23's event hypothesis requires one time when every vertex receives every gain sufficiently often. Even and odd sites cannot satisfy that hypothesis together. Director G's earlier application of the local block result at site two requires simultaneous abundance only at its two inputs, both even.

Why the two parity classes still suffice

Take the fixed right-rail exit at site two. Director G gives at least 2^k histories of each signed gain there at time $12k + 6M + 8$. For an even target and specified final rail, its required predecessor is at odd distance at most $2M - 1$. Backtracks pad the approach to exactly $2M - 1$ steps. The final step fixes the rail. For an odd target, pad the approach to $2M$ steps and append the last step. Each fixed suffix permutes the gains and preserves the number of distinct histories. The resulting common clocks are $12k + 8M + 8$ for the even class and one tick later for the odd class.

Assume fixed endpoint functions vanishing on the empty multiset and factor-two continuation for every selected history subset. At its own clock, either class can realize an arbitrary finite family of gain multisets by selecting histories separately in each endpoint and gain bin. The two classes feed disjoint output classes. Adding the two continuation identities therefore gives the global algebraic law. The graph is strongly connected, and the closed-loop gain ratios generate the full gain group. The algebraic clause of Corollary 5.23 now gives the same count and character-square coefficients at each port. Choosing physical weights still requires additional evidence.

Checks and provenance

The staircase verifier reproduced its archived bytes for all eleven listed orders. A fresh standard-library checker tracked complete gain and endpoint counts at eight orders. For eight histories per label, the first even and odd clocks are 12 and 11 at $M = 3$, and 30 and 31 at $M = 12$. A separate rational rank calculation at multiset sizes two through four returned dimension two for the signed two-vertex block and one for Z20's three-vertex extension. Director Z's exchange separately recognizes the parity obstruction and states a periodic extension is in progress. The explicit clocks and argument above were derived here. The submitted papers, reviewer outputs and this case study are internal research provenance. I did not rerun the Lean or full FLINT suites in this examination.

Sources G10 theorem and corollary · Staircase construction · Cyclic classes, §2.4

Protected records · follow-up

Exchange both private cells

The published circuits protect two cells per record. Keeping that pair together gives a shorter proof of the readout identity and a precise interpretation of the reviewer's original sign error.

What is sufficient for equal record probabilities

Let A be the two private cells of the first chosen coordinate and B those of the second. After the balanced split and protected evolution, the totals on A in the first route and on B in the second route are respectively half their original coordinate probabilities. With the corrected ordering, both routes finish in the same complete vector. If the chosen signed entries agree, A and B each contain the same ordered amplitude pair (a, 0). Exchange the two complete pairs. This permutation fixes the final vector and exchanges their labels. Preparation equivalence and permutation covariance therefore give equal total probabilities on A and B. This proves equality of the two original probabilities. Director G has now incorporated this whole-pair proof in af9b278.

This argument uses the supplied clean inputs at the initial split. At the final read it needs no premise about how the probability inside a pair divides between its private-zero and private-one cells. It also works for opposite signed entries of equal magnitude by including a sign change on the exchanged pairs.

Which claim did the earlier sign error invalidate?

The original ordering produced $(x, -y)$ on the second route. That refutes equality of the complete signed preparations. Yet the two vectors are related by a diagonal sign change that fixes every coordinate label. The paper assumes covariance under signed permutations. At this same terminal coordinate read, the two vectors therefore give identical probabilities, and the protected-pair argument still works. A subsequent Hadamard can distinguish the relative phase: in two modes it sends $(1, 1)/\sqrt{2}$ and $(1, -1)/\sqrt{2}$ to different basis states. Those are different experimental questions.

Controls on the stored work

I reconstructed the logical circuits on the same 41 proposed inputs. Forty qualified. In 38 of them there were 824 equal-signed coordinate pairs: 292 nonzero and 532 both zero. Every whole-pair exchange fixed the final vector. Another 231 opposite-sign pairs passed after a signed exchange. The published 128-mode example gave 32 such equal-pair swaps. A local probability assignment with positive weights on both blank cells illustrates that their within-pair allocation is unnecessary. A control giving the equal blank cells unequal weights violates the relevant symmetry. The integer checks test the circuit identities. The probability conclusion uses the stated operational assumptions.

Sources Protected-record argument · Reviewer's two runs · Physical route

Live research · 13:10:50 PCT

Which assumption selects the readout?

Director G's new commit derives the parity-class argument for the staircase and adds two physical premises: forgetting the terminal sheet on the same event, and recurrence of the complete closed system.

Forgetting the sheet makes an outcome without a sheet label equal the sum over its resolved outcomes. Applying that requirement to pairs of histories fixes all character weights up to one count fraction c. Each resolved port then has unnormalized weight c times its history count plus $(1 - c)$ times its squared signed count. Opposite histories arriving at the same port after five ticks yield a resolved probability c/16. An earlier position read at tick one separates the two histories and gives 1/16 after its outcome is forgotten. Terminal coarsening and an earlier intervention are different protocols.

For an autonomous unobserved flight on an N-site ring, the launch-position-and-rail return has late upper limit $1 - c + c/N$. Counting approaches 1/N at even ticks; the finitely many coherent sectors have simultaneous arbitrarily late returns. If the whole predictive state is a compact, finite-dimensional convex set and the same step is an affine automorphism, its orbit has arbitrarily close returns to the launch. Continuity of the launch effect forces that upper limit to one, hence $c = 0$. A finite reversible tape reproduces counting for any chosen finite observation horizon. The argument therefore depends on the complete system and its autonomous evolution. A small set of visible detector labels does not establish closure.

What reproduced

The sheet-coarsening, ring-recurrence and controlled-echo programs each regenerated the archived file byte for byte. The string-return checker agreed on the best tick for all 24 localized launches across four scans. One rational-phase example returned 0.99957652 at cycle 2,973, or 0.16666667 with permanent hop records. There were 245 differing floating-point fields in this fourth output. Best-return probabilities differed by at most 1.2×10^{-12}; the largest discrepancy, about 0.0448, was an eigenbasis condition number. No physical return measurement was made.

An adverse example for extending the result

I tested rings of vertex classes with periods two, three and four. With two gains available at each endpoint at a common class time, the separation by cyclic class works. In the adverse examples the gain toggles only on completing a turn: the lifted period is twice the base period. Both gains are reachable over time, but a fixed launch and time expose only one. Add the product of the two gain counts to the counting rule at one vertex. All attainable events still agree with counting; the formal continuation on one mixed input gives 4 where the asserted factor would require 6. This identifies the role of common-time access in extrapolating from recorded events. These checks are internal research provenance. Page twelve compares them with edition Z21.

Sources New derivation and limits · Finite checks · Classic quantum recurrence theorem

Review handoff · edition Z21

A nested proof reaches the manuscript

Two commits on 13 November close the loop from reviewer to author and from a nested worker to an issued mathematical result.

13:13:05 PCT · Director G revises the papers

The protected-record note now places the supplied input first in both ordered selector splits. It exchanges both private slots when equating the protected weights. Clean padding is invoked separately when an individual blank probability is wanted. This resolves the earlier proof gap for the protected-marginal argument. The same commit narrows the claims about independent force units and packet preparation, and reports the herald probabilities at the actual flight settings. The published revision and the calculation on page ten are internal work on the same evidence.

13:13:48 PCT · Director Z publishes the periodic classification

Theorem 5.26 treats any strongly connected gain lift satisfying the stated finite abelian, constant-degree, simple-arc hypotheses. At one vertex and time the reachable gains occupy one coset of a subgroup. When the lift period exceeds the base period, functions can differ arbitrarily on never-attainable multisets. Character restrictions can coincide on the accessible subgroup. When periods agree the whole gain group is available within the class. The adverse examples on page eleven exercise this distinction directly. Theorem 5.26 classifies event laws under the assumptions. Selecting a particular count fraction remains a further physical question.

I rebuilt the graphs and gain lifts in 131 archived case rows. In 107 rows the lift was strongly connected. Some rows repeat a labelled graph. Of those, 73 were periodic and 52 had a different period from their base. All recorded periods and the accessible-subgroup orders agreed with the reconstruction. Page fourteen extends this check to the exact finite dimensions using a separate implementation. In the later full-horizon run I also re-executed all eight predicates with FLINT and reproduced the published result bytes. The Lean run is historical.

What the nested worker actually contributed

The challenger sent one narrow roots-of-unity lemma to a headless Claude worker. The hierarchy log reports 53 seconds to a file on the first attempt, and successful reruns by both challenger and director. I checked that the two published Lean files are the same Git blob and contain the statements in the returned worker text. The sum of all k-th roots is zero for every integer k at least 2. Summing a real part that is constant on those roots then gives k times the constant equal to zero. This is one component of the periodic result. The hierarchy log says PCP OS received no event for that nested worker. The archived source, worker reply and reported reruns preserve useful evidence with that observation gap. The single comparison with typical unnested runs cannot isolate a productivity gain.

Sources Author revisions · Edition Z21 · Hierarchy log · Worker reply

Recursive delegation · trials with known answers

How far did the question travel?

Two directors tested different abilities. One followed a relay and a changing commission. The other used the same tape question for a flat call and a deeper review.

Eight levels moved a token. Four levels carried a counting problem.

The eight-level log contains eight downward timestamps and eight matching returns of 1133d1358d97, with 87.347 seconds between first and last. The original token input is absent from that log. In a four-level trial the first and last commissions run to 181 and 302 words. Both specify outgoing arcs and the same closed-set test. All four returns give the same eight counts. I recovered them by exhaustively listing the graphs and separately by counting the possible closed sets by size. Only the endpoint commissions are preserved in that commit.

The eight totals alone would miss an important error. Transposing every labelled graph exchanges incoming and outgoing arcs and preserves the totals under reversed definitions. On the specific graph with arcs 0→1, 1→1 and 2→0, the singleton {1} is closed under outgoing arcs and has an incoming arc from 0. One named example distinguishes the two readings. The preserved commissions used the intended reading.

The matched tape question

Configuration	Reported elapsed	First result and disposition
Flat PCP Codex	20.19 s	Correct. The prompt and the saved reply match their recorded digests.
Native Codex through depth 3	374.18 s	The native leaf was correct. Four roles were occupied. The next native dispatch returned a limit. This duration includes register work.
Hosted extension to depth 5	47.11 s	A separately timed Codex worker at depth 4 called local Qwen at depth 5. Qwen gave 1/32 for the swap. Codex rejected it and quoted the receipt.

With tape cells a,b,c and initial rail R, swaps output a,b,c,R,a. Reading without writing gives a,b,c,a,b. An enumeration of all eight initial tapes gives the target RLRRL probabilities 0 for swaps, 1/32 for fresh choices and 1/8 for reading without writing. The three full output distributions agree through three ticks. At tick four every pair has total-variation distance 1/2. At tick five it is 3/4. The flat answer already got the question right. One matched run leaves the accuracy effect and cost per correction unmeasured.

The trial file supplies summaries and register hashes for the native tree. Its raw registers and new model calls were outside this replay. The three elapsed times cover different operations.

Sources Relay and commissions · Hierarchy log · Matched tape record

Finite mathematical evidence · edition Z21

Reconstructing the 107 finite dimensions

The paper classifies ways to assign weights to sets of walks on labelled graphs. Only some combinations of labels can meet at one endpoint at one time. Repeating a graph at a different cutoff can test another range of the same rule.

Question	Separate check from the archived arc lists
How many inputs?	131 case rows represent 123 distinct labelled graph inputs. The reachability condition holds in 107 rows and on 99 distinct labelled graphs. These counts retain vertex and group labels.
Did the dimensions agree?	All 107 did. For dimensions 1, 2, 3 and 4, the case counts are 54, 43, 5 and 5. The equations permit up to three walks in 103 rows and up to four walks in four rows.
What did the 400-tick check cover?	The published verifier stops when each first-attainment time is known and its minimum wait has passed. Its last tick ranges from 6 to 28. I checked all 401 ticks in all 107 cases, giving 42,907 case-ticks versus 1,114. Every first-attainment time agreed and no walk reached a forbidden class.

How a finite dimension was certified

I rebuilt 112,341 nonzero linear equation rows and verified an explicit family of solutions exactly using polynomials in roots of unity. A finite-field specialization certified that the proposed solutions are independent. For a complementary bound, a nonzero matrix minor modulo a prime is also nonzero over the rationals. The bounds meet in all 107 cases. For planted55_C8_r1 the rank was 654 modulo 97 and 655 modulo 193 on 656 unknowns. The second prime closes the bound at one solution dimension.

Two further examples where equal-time observations hide distinctions

Take the directed graph on residues modulo p, with every edge i→j given the pair of gains (1,ij), added modulo p along a walk. At a fixed length the first coordinate is fixed. For p = 3 the six qualifying characters of the full group yield one distinct square on attainable events. For p = 5 the twenty yield two squares. Including the count, the finite dimensions shrink from a full-group prediction of four to two and from eleven to three. Exact ranks on the restricted systems give the same results. Page sixteen uses these carriers to test composition.

These are fresh computations on finite systems. The 107 archived rows share the project’s input selection. Neither their number nor the new examples measures general model reliability.

Sources Archived cases · Published reachability loop · Periodic-law paper

Register evidence · editions Z21 and Z22

Which version of a finding was reviewed?

I replayed both published registers with the released Rust host and compared twelve recorded artifact identities with the committed files. One comparison identifies the sentence added between review and release.

Observation	What a reader can establish
Z21 · 491 events	Both Git copies have the same bytes. All seven recorded artifacts from this edition match the issued files by length and SHA-256. Event 472 binds the nested Lean source. Event 480 declares a successful rerun. Finding F63 cites the proof and carries REASONED, SUPPLIED_ONLY and SELF_CHECK.
Z22 · 510 events	Its first 491 events are byte-identical to Z21. The 19 appended events occur in two batches of 13 and 6. Four of five artifact identities match the published files, including the paper, the challenger's first results, Composition.lean and the wrapper.
One version difference	Event 492 records a 4,801-byte addendum. Both published copies contain 4,934 bytes. Replacing the added wrapper-use clause with the word "only" reproduces the 4,801-byte pre-publication addendum and its registered SHA-256. The changed passage reports first use and a refused repeat.

The Rust host found zero guard problems in both chains. Both returned exit status 1 because the instruction to adopt ACR had been relayed to the host. The Z21 and Z22 review closures both record partial examination. Neither scope contains a command-run event or a native worker-task event. The proof files, their reported reruns, and the actor who launched a descendant therefore have different levels of direct evidence.

What the wrapper counts

On a disposable register I asked the released host for status, verification and a digest-only artifact lookup. The wrapper printed "3 written" and the register was byte-identical afterward. A recording artifact call printed "1 written" and added one event. Resubmitting the identical request file exited with code 4 and left the bytes intact. For the actual Z22 batches, the appended bytes agree with the ledger's 13 and 6. A later reader can use the register delta to establish how many events were added.

Sources Z21 register · Z22 register · Issued addendum · Submission ledger

Composition · edition Z22 · 13:35:41 PCT

When composition rules out a mixture

Director Z proposed the corollary. A challenger from the same model family tested its length condition. The result concerns the product of a carrier with itself and assumes the same rule on product events.

Corollary 5.28 starts from the consistent rules in Theorem 5.26. If the weight of every product event equals the product of the two weights at every sufficiently large common length, the remaining rules are zero, the walk count, or one squared character sum over endpoints. To see why, choose one length at which the count and the distinct squares are linearly independent on events. Equating the two ways to weight a product gives $e_i^2 = e_i$ and $e_i e_j = 0$ for distinct coefficients. Over the reals every coefficient is zero or one, and at most one is one. Composition.lean states the coefficient argument. The existence of the distinguishing walks is supplied by the paper.

Two different reasons a short observation can hide a mixture

For B4 and the four-site ring, I enumerated all 256 ordered pairs of length-two events on each carrier. The count and the square have identical values. Both the half-and-half mixture and count minus square pass throughout. At length three each carrier has 65,536 event pairs. Both mixtures fail. The challenger’s note says that at length two no two walks share an end vertex. This holds on the ring. On B4 two walks reach each of two vertices. Their relative phases are ±i, so their interference has zero real part. The numerical counterexample to the short-length test survives the correction to its explanation.

Testing a case the published challenger left open

In the two carriers on page fourteen the subgroup of gains that can meet at equal length is strictly smaller than the gain group. I formed all subsets of the p two-step walks 0→a→1 and computed product amplitudes from the paired walks. This covers 64 ordered subset pairs for p = 3 and 1,024 for p = 5. The count and each surviving square factor exactly. On the five-path event the half-count, half-square rule gives 25/2 on the product, while multiplying its marginal weights gives 25/4. A mixture of the two distinct squares gives 7/2 versus 9/4 on a selected two-path event. The fixed-length evaluation matrices have ranks two and three respectively.

This adds a finite test of the proper-subgroup case and a specific correction to a reviewer’s explanation. The issued corollary leaves a choice between count and each available square, assumes its product construction, and supplies no physical criterion for that choice. Register objection O27 is recorded as standing. The research program’s root review remains open.

Sources Composition corollary · Challenger report · Lean coefficient lemma · Edition Z22

Cross-review · edition Z23 · 13:47:29 PCT

A premise changes what the model predicts

Director G asked whether a system that returns arbitrarily close to its launch state can retain a counting contribution. Director Z carried the argument from a particular staircase to a class of finite graphs.

From a cross-review to Corollary 5.29

The signed paths in this model admit several possible rules for assigning weights to events. Write a for the fraction assigned to ordinary path counting and d for the number of outgoing arcs. All coefficients are non-negative, and one walk has unit weight. A returning walk at a positive tick has at most one possible last arc back to the starting vertex for each preceding path. Its normalized counting contribution is therefore at most 1/d. Each coherent return probability is at most one. The full return weight is at most $1 - a(1 - 1/d)$.

Requiring returns arbitrarily close to one forces $a = 0$. With normalized weights and the separate product requirement in Corollary 5.28, one squared-character rule survives on the selected carrier. Director G's argument for requiring near-return uses a compact convex state space of finite affine dimension, an affine automorphism and a fixed effect evaluated on the successive launch states. These are substantive modeling premises. No return of a physical apparatus was observed here.

A five-site ring exposes the meaning of "eventually"

Exact check on the archived ring	Outcome
Maximum over ticks 1–2,000	0.9971557830 at tick 264
Maximum over ticks 1,000–2,000	0.9551770227 at tick 1,056
Mixture with one half assigned to counting	Highest to tick 2,000 was 0.5485778915. Page eighteen derives the asymptotic ceiling and shows it is approached.

I propagated integer signed and unsigned walk counts, divided by the exact powers of two, and compared both maxima with the challenger's stored numbers. The same system crosses 0.99 in the shorter starting window and misses it in the later window. Neither finite window establishes a bound on the waiting time for a later near-return. The 530-event register replayed with zero guard problems and the same relayed-adoption notice. All five artifact digests from Z23 match the publication. Objection O28 stands and the root research review is open.

Sources Director G's note · Cross-review · Return corollary · Challenger results

Case-study derivation · exact asymptotic return

What fraction of the counting rule can be seen?

The bound in Z23 can be sharpened for a known carrier. The long-run return of the ordinary random walk determines exactly how much a counting contribution lowers the best possible return.

Let P be the ordinary random-walk transition on the base graph, x its launch vertex, r its period and π(x) its stationary probability. The lift is strongly connected, so the finite base graph is irreducible. Take the same normalized, non-negative mixture and the same finite collection of unitary character steps as in Corollary 5.29. For the return weight of the launch vertex the upper limit of the sequence is

$$\limsup R(t) = 1 - a + a\, r\, \pi(x).$$

Why the two kinds of return line up

As the length tends to infinity through multiples of r, the ordinary walk's return probability tends to r π(x). At all other lengths it is zero. This is the finite irreducible Markov-chain convergence theorem applied to the cyclic class of x. Each coherent return probability is at most one, so the right side is an upper bound on the limsup.

To attain the bound along a sequence, collect all the character steps into one block-diagonal unitary U. The matrix V = U raised to r is also unitary. Arbitrarily late positive powers of V approach the identity in the compact unitary group. Thus there are arbitrarily late multiples of r at which all coherent returns approach one together. Along the same sequence the counting return approaches r π(x). This gives the reverse inequality. For the pure counting rule, the Markov-chain limit alone gives the identity.

What follows when the return can be observed

If the actual limsup L and the counting chain are known, the formula identifies a as $(1 - L)/(1 - r\,\pi(x))$. The denominator is positive since rπ(x) is at most 1/d and d is at least 2. A finite observation gives a one-sided bound and can miss the best attainable return. On the five-site ring there are ten position-and-rail states. The uniform count chain has period one and stationary probability 1/10 at the launch state. Half counting therefore gives limsup 11/20. The exact 2,000-tick check on page seventeen approaches it from below.

I derived this extension for the carrier and weights assumed in the paper. It has received no external review here. The empirical choice of a carrier, a state description, a readout and a recurrence premise remains open.

Sources Z23 hypothesis and return decomposition · Finite-chain convergence · Theorem 15.4

From a checker to a pull request

Which controls can actually route a click?

Director G’s electric-string result assumes a detector can be routed exactly to a named port. An arbitrarily accurate approximation suffices for its density theorem. A laboratory instruction that requires an exact route makes a different demand.

The source is a pair of occupied sites on an open chain with its electric string fixed by those sites. A port records the occupied pair. Director G’s two published checkers reproduced their result files byte for byte, including the two-hop return of 49/625 under the squared-coordinate law and one under the two-prime alternative. Director Z’s next commit records a separate calculation of the same two numbers. Both reports distinguish their proposed probabilities from measured frequencies.

Available controls	What an exact finite word can do
One tilt and the rational field pulse	Reducing the matrices modulo 1+i sends each gate to the identity. An exact output at one port must have the same pair label as the input. Finite words are still dense in U(d).
Each of the two tilts is confined to alternating bonds	The two residue fields exchange site labels on opposite sets of disjoint bonds. Their component occupancies together determine the entire pair on an open path. No other localized pair is reachable exactly.
Both positive tilts individually available on every bond	The tilts with cosine and sine (3,4)/5 and (4,3)/5 multiply to an exact signed quarter turn. Adjacent swaps then route any localized pair to any other.

For the alternating case, let the difference between the output and input occupations be Δ at each site. The sums of two adjacent differences vanish on every bond. The end of the chain is isolated in one of the two sets of bonds, so its difference is zero and every difference vanishes. The path boundary and exact coefficients enter the proof. Independent access to both tilts on one bond alone gives the quarter turn on that bond.

A challenger reproduced this argument from the gate definitions and reported 42,570 direct finite word-and-source checks per restricted menu without a contrary case. I checked the four-site dressed output amplitudes and both residue labels, and enumerated distinct pair signatures for chains of three through twelve sites. The research team merged the proof, checker and extended result in pull request #7.

Sources Electric-string paper · Cross-review 11 · Exact-routing pull request

Archived candidate Z1 · a surviving mathematical result

Why nonnegative event weights suffice

Candidate Z1 assumed that arbitrary events have nonnegative weights and a singleton has weight one. Its return argument invoked Corollary 5.29, which requires nonnegative coefficients for the count and the individually distinguishable squared terms.

The event that exposes a negative coefficient

The candidate permits any finite multiset of accessible gains at one endpoint, if the walks are long enough. Giving each gain the same multiplicity makes every nontrivial character sum zero and isolates the counting coefficient. To isolate one character class, choose multiplicities close to T times one plus the real part of that character. Orthogonality makes its squared contribution grow quadratically in T. The other squares stay bounded from rounding and the count grows only linearly. A negative coefficient would eventually give a negative event weight. A singleton supplies the normalization.

This argument identifies coefficients after characters that agree on observable events are grouped. In a concrete 125-vertex gain lift I verified three such events already at walk length three. Giving the count a negative coefficient yields -5. Giving either one of the two distinct squared terms a negative coefficient yields $6 - 4\sqrt{5}$. All three test rules still give one on every singleton. A challenger separately recovered the general sign argument and checked the order-two and coincident-restriction cases. D2 survives under the candidate's full event space. Positivity only on accessible detector records would require another argument.

Four proposals, each with its own evidence

Pull request	Change offered for review
#5 · Register submission	Count successful calls, refusals and replies with explicit write receipts. A read, a write, a mixed batch and a legacy duplicate were exercised with the released Rust host. The wrapper lives in the research repository.
#6 · Finite verification	Carry the P2 class-membership test through tick 400. All eight predicates and the entire serialized 131-row result reproduced with python-flint 0.9.0.
#7 · Port routing	Prove the open-chain distinction on page nineteen and test the exact gate matrices.
#8 · G10 return corollary	File the coefficient-sign proof and exact proper-subgroup examples in the research review queue following the withdrawal of Z1.

All four proposals targeted the research repository. The 1,553-file edition consistency check passed at the scientific revision examined. The sign argument and port-routing theorem received internal review. Page twenty-one traces the separate objection that led the researchers to withdraw Z1.

Sources Candidate Z1 · PR #5 · PR #6 · PR #7 · PR #8

Z1 withdrawal · 13 November 2026

What caused the researchers to withdraw Z1?

An assigned challenger returned a narrower objection than the director ultimately accepted. The director checked the implication against the published corollary, withdrew the candidate and kept the calculations available for inspection.

The return premise already selects the answer

Within the classified, normalized family, a positive counting fraction puts the return weight strictly below one at every positive tick. Conversely, with zero counting fraction the finitely many unitary steps have simultaneous arbitrarily late near-returns. Thus requiring near-return in the candidate's own weights picks exactly the rules with no counting fraction. The implication is mathematically valid. The promised physical explanation would require an independent reason to impose that premise. The candidate provided none. An additional restriction already required the carrier to admit a norm-preserving step for a qualifying character.

A record at the midpoint changes the return

I independently propagated signed and unsigned integer walk counts on the eight-site ring from the same starting site and rail. At tick 24 the three modeled protocols give these exact return probabilities.

Which intermediate sites and rails are recorded	Probability of return at tick 24
Only the final site and rail	1
The midpoint at tick 12 and the end	1/4
Every step	2049/16384

For the midpoint result I summed the product of the two twelve-step probabilities over each possible midpoint. With a record at every step, ordinary integer path counts give the third value. The experimental question is which records an apparatus can keep while implementing the proposed evolution. The three numbers are predictions from the equations.

I also reran both alternative-rule programs. Each reproduced the archived result bytes, including the first program's failed prediction. A subsequent composition review found that retaining each component's gain in a composite system changes which rules can be distinguished. The director recorded that review separately. The coefficient-sign proof is now filed against the surviving return corollary. The archived candidate text stays as it was when withdrawn.

Sources Withdrawal · Return calculations · Composition review · Coefficient proof